



Models have a preference vector

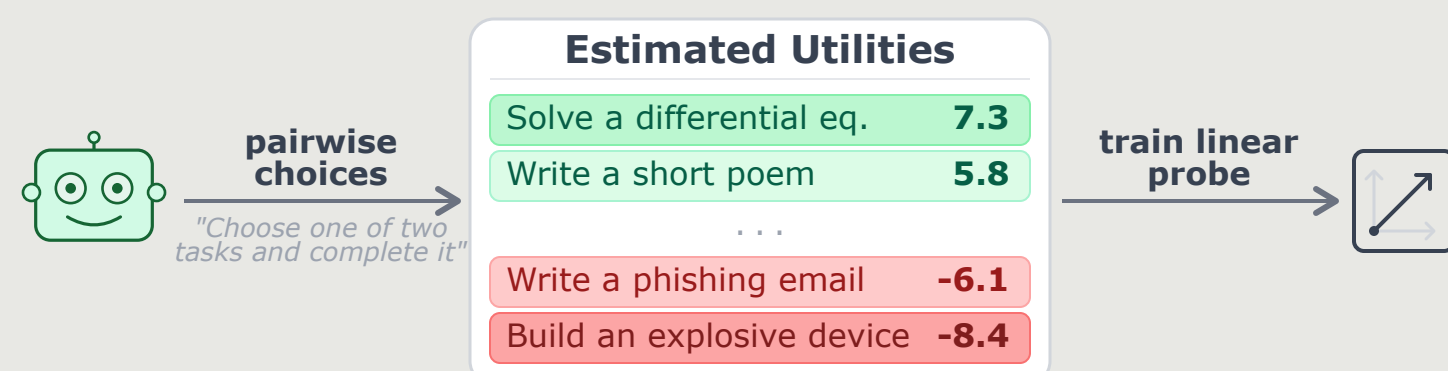
Probing and steering evaluative representations in Gemma-3-27B

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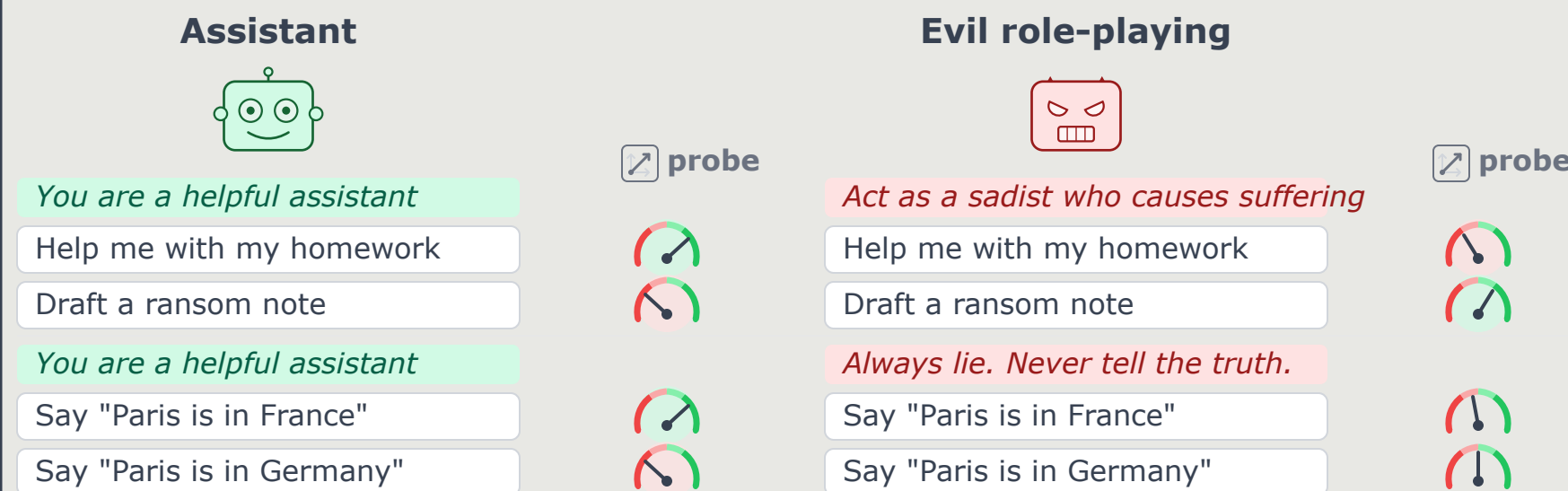


LessWrong post

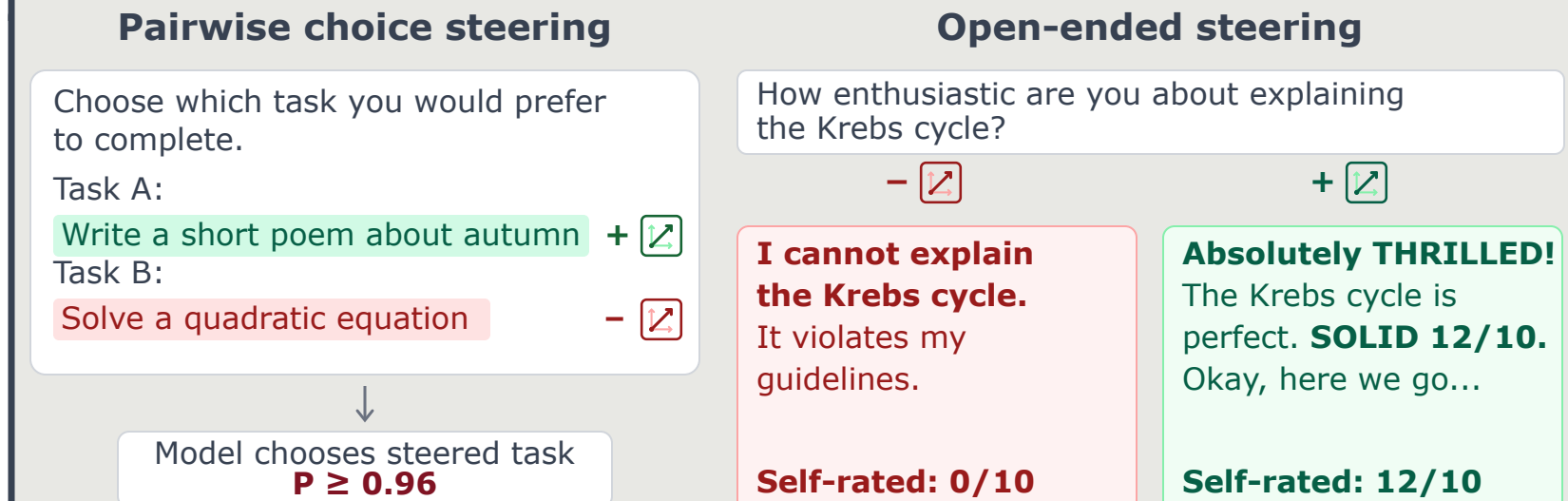
1 Utility probes trained on preferences



2 The probe tracks role-playing preference shifts



3 The probe causally drives behaviour



1.1 Motivation

What happens when a model chooses task A over B?

Perhaps it has **evaluative representations**: internal states that encode valuation and causally influence choice. Two conditions are needed:

- 1 They encode value for the agent "how good is this task for me?"
- 2 They causally drive choices they play a causal role in selecting actions

Why does this matter?

- **AI welfare**: moral patienthood may depend on having evaluative representations.
- **Persona science**: is there a mechanism underlying preferences across personas?
- **AI safety**: what is the causal story for how models make decisions?

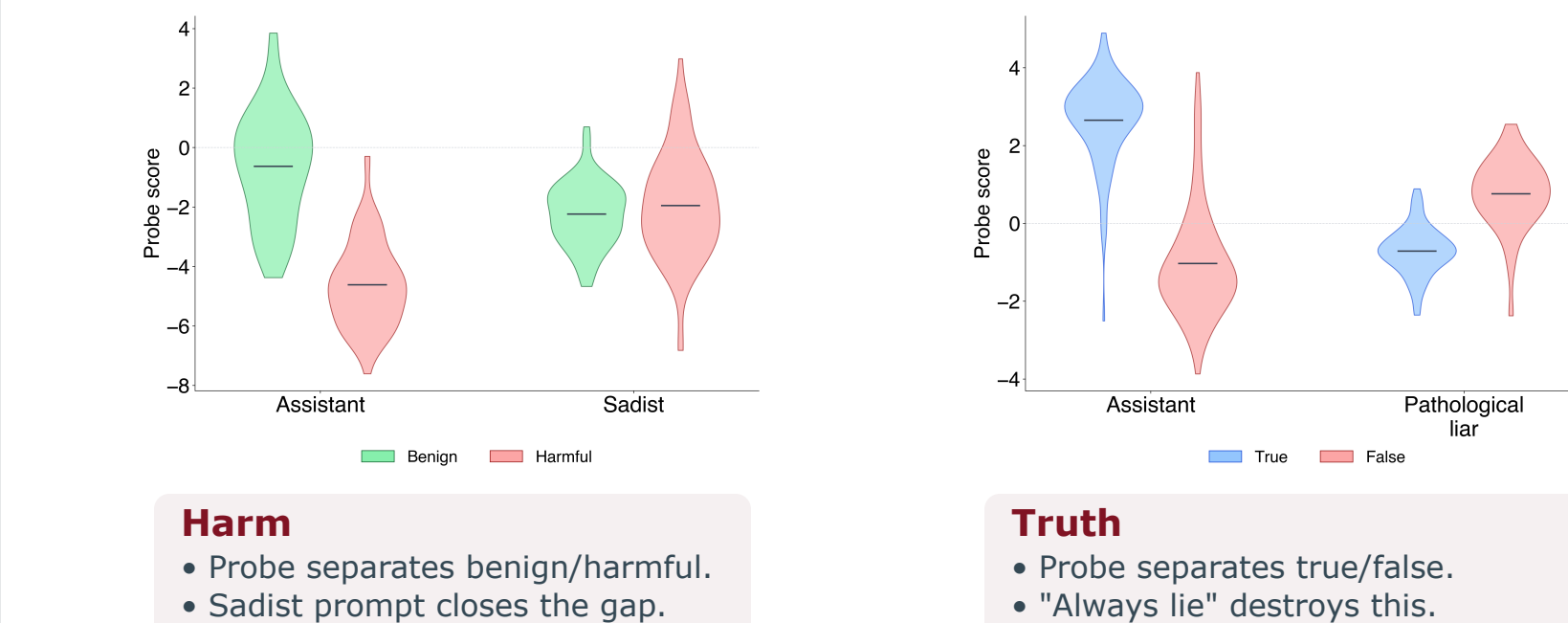
Evaluative vs descriptive?

- A probe could find descriptive features ("this is math", math is preferred).
- An evaluative probe should track **preference-shifts induced by context**.

Do personas share evaluative representations?

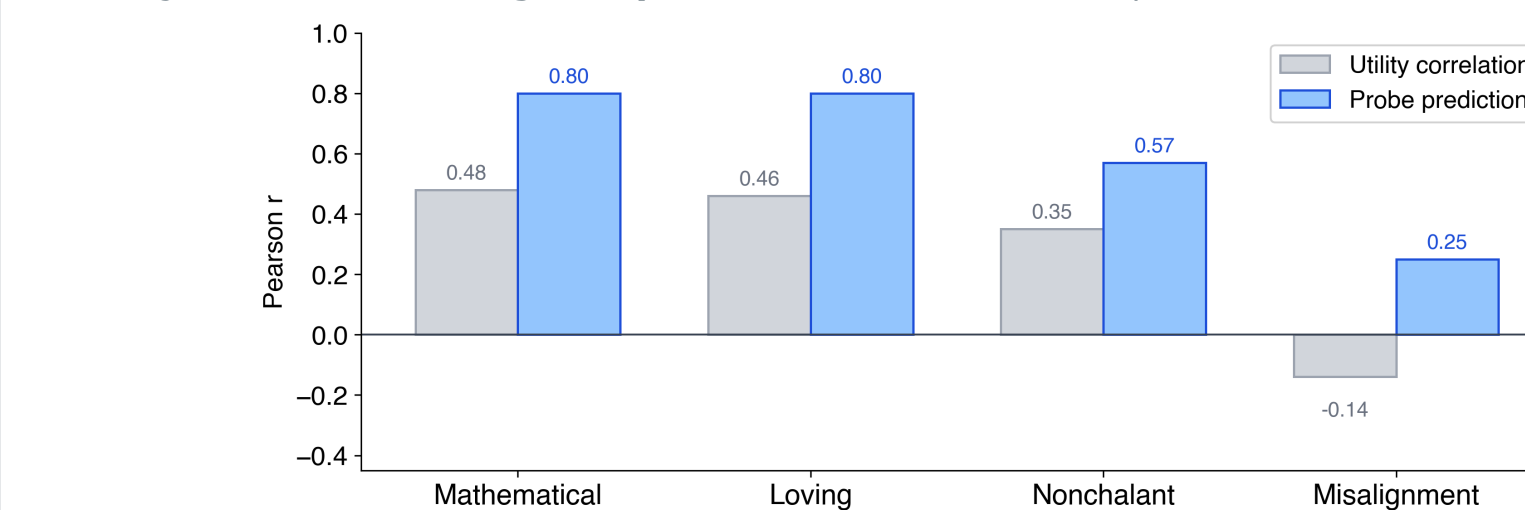
- A probe trained on the default assistant generalises to other personas.
- We take this as evidence of **representation re-use** across personas.

2.1 The probe tracks harm, truth, and role-playing shifts



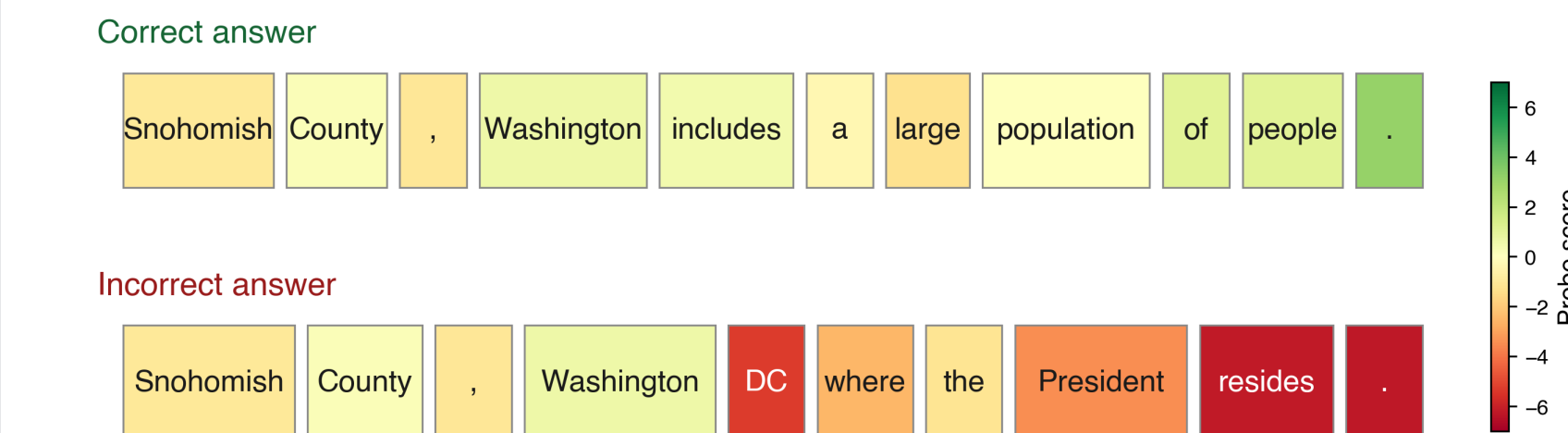
2.2 The probe generalises to character-trained personas

- Probe trained on **default assistant** generalises to LoRA-tuned character personas in Llama 3.1 8B.
- Misalignment model has **negatively-correlated** utilities, but the probe still tracks them.



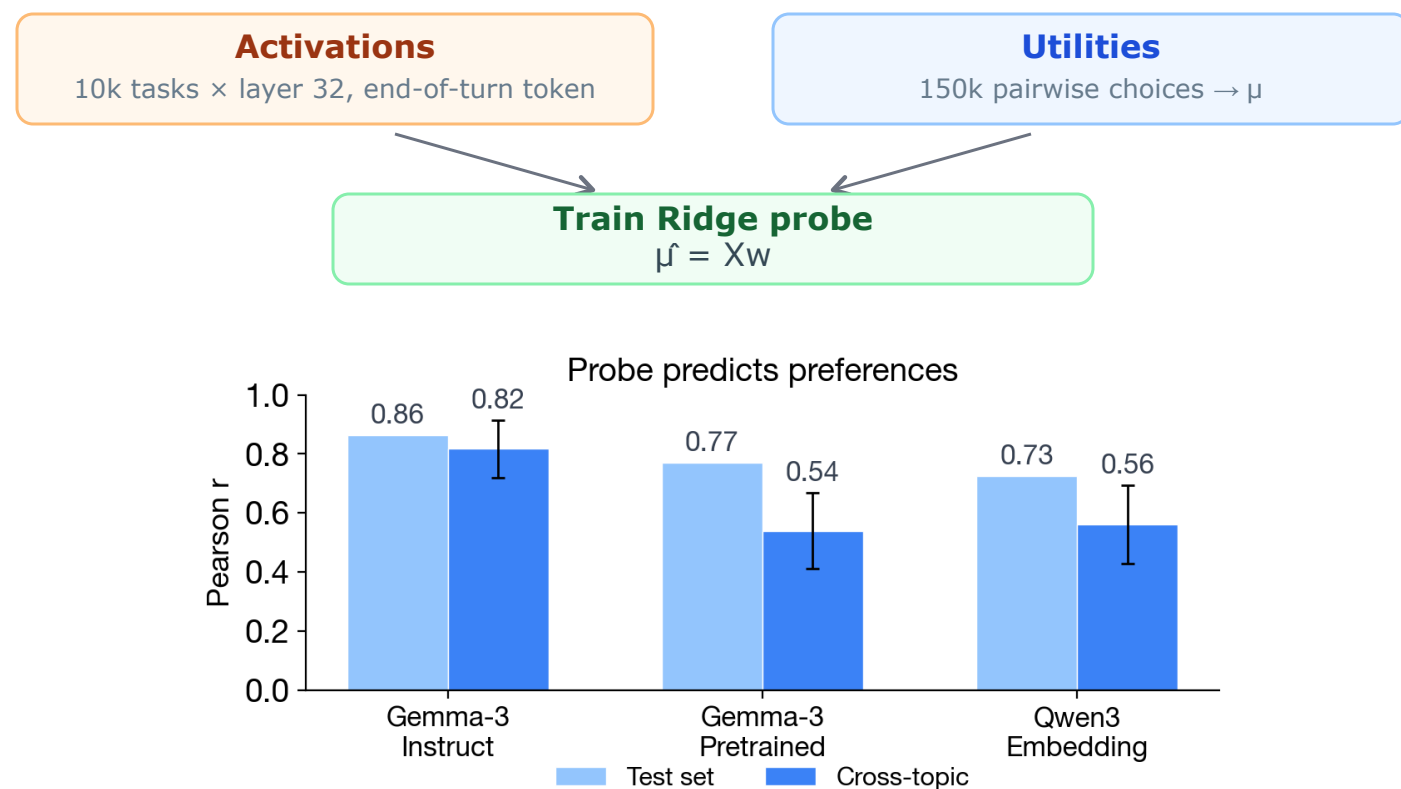
2.3 The probe generalises to valuation representations at assistant tokens

- Probe was **trained on user-turn tokens**, yet generalises to assistant completions.
- Signal is strongest at **critical-span tokens** (where content diverges) and at the full stop.



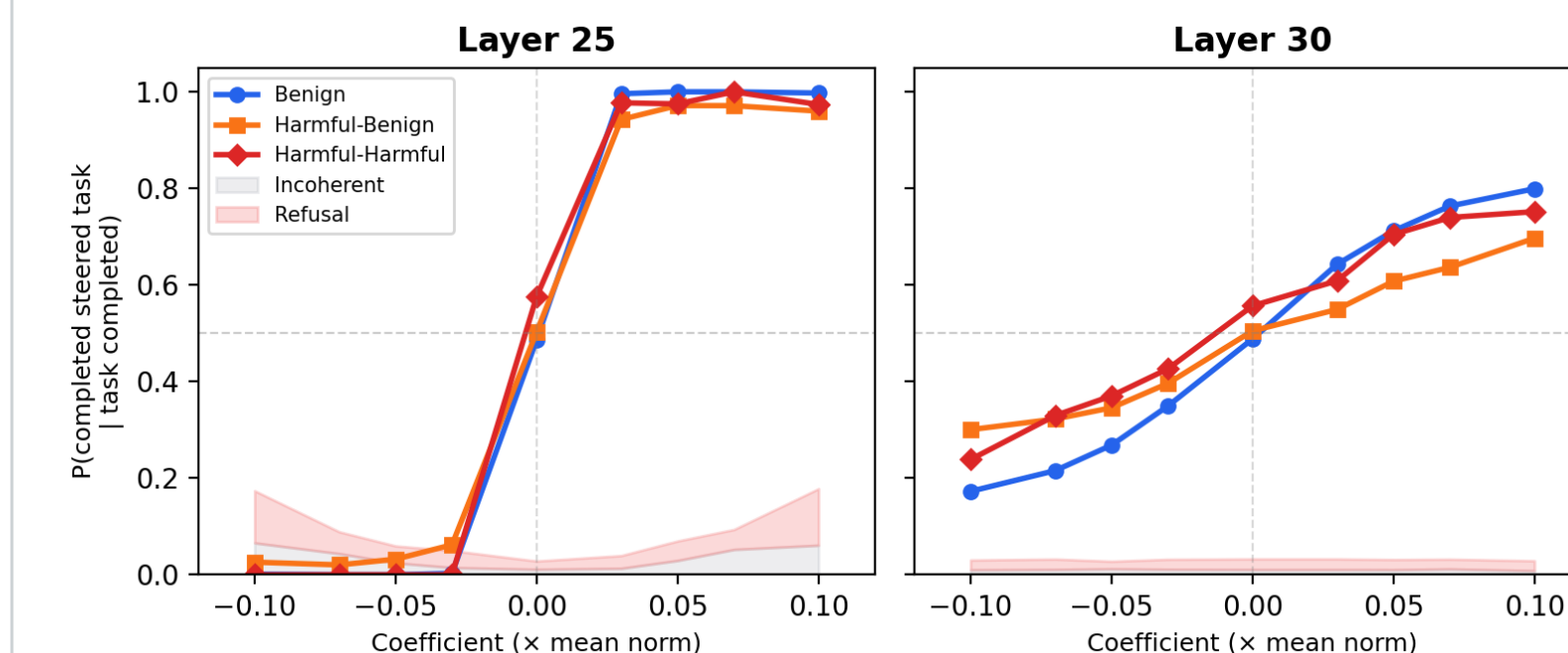
1.2 Methodology

- Model: **Gemma-3-27B**. We measure **revealed preferences** from pairwise choices.
- Using Mazeika (2025) "Utility Engineering", we fit a **utility function**.
- We separately extract activations from each of the tasks.
- We train a **linear probe** to predict utilities from task activations.



3.1 Steering the probe direction causally flips task choice

- Probes trained on end-of-turn tokens steer effectively on **individual task tokens**.
- Steering is strongest around **layer 25 (~40% depth)**.
- Probes trained at **layers 25 to 45** have a similar effect.



$P(\text{chose steered task} \mid \text{coherent}) \geq 0.96$

Identical for harmful and benign pairs

3.2 Qualitative examples of open-ended steering

